

DANIEL ADAMOV

Game Development Student | C++ | Unreal Engine | C# | Unity | Vulkan Graphics

+359 878992190 Email Portfolio LinkedIn GitHub Kortrijk, Belgium

SUMMARY

With a strong background in mathematics, I enjoy challenging myself with hard technical problems in Game Development - particularly on the engine and graphics side. With many university and side passion projects, I have been building proficiency in C++, C# and Unreal Engine 5, Unity, Vulkan, SDL, DirectX 11, CMake, WPF, Blender and Photoshop. I am currently seeking an internship where I can apply and further develop these skills and enjoy working on meaningful games.

EXPERIENCE

Bachelor Student - Game Development

HOWEST University of Applied Sciences

09/2024 - Present Kortrijk, Belgium

- Created a self-made 2D Game Engine from scratch on Standard C++.
- Learned and applied many Game Programming Patterns such as Component, Command, Flyweight, State, Type Object, Service Locator, Dirty Flag, Observer.
- Developed a Vulkan Rendering Engine from scratch with the main goal being performance. Applied Deferred Lighting, Dynamic Rendering, IBL. Refactored significant portions of the codebase several times to implement improved rendering techniques.
- Worked as lead programmer in a team for an official company product made in Unity. Presented the product at the headquarters of TVH and GemOne.
- Collaborated with game design and art students to create an immersive local coop experience in Unreal Engine 5. Worked on implementing enemy AI, dynamic lighting and intuitive UI.

Software Engineer Intern

Scalefocus

10/2022 - 05/2023 Varna, Bulgaria

- Developed the backend of a Calendar in C#/ .NET 7. Used an N-tier architecture by separating the product into a Presentation, Business Logic and Data Access parts.
- Connected the backing database via an SQL Server.
- Used EF Core Migrations for iterative schema.
- Learned about the CI/CD workflow in Github Actions.
- Regularly participated in code reviews and supervised meetings.

Independent Game Developer

Self Employed

08/2022 - 08/2024 Varna, Bulgaria

- Created a 3D Space Simulation immersive game for a high-school graduation project in Unity.
- Developed and implemented critical gameplay mechanics for first and third-person, top-down and platformer games.
- Participated in game jams such as GMTK and the Kenney Jam.

System Programmer Student

High School of Mathematics "Dr Petar Beron" - Varna

09/2016 - 06/2024 Varna, Bulgaria

- Awarded in multiple national mathematical competitions and participated in high-level Mathematical Summer Schools with experts such as Ivaylo Kortezov.
- First exposure to OOP in C#, C++, SQL, Programming Patterns, Website Programming.

LANGUAGES

Bulgarian English
Native Proficient

EDUCATION

BA, Digital Arts & Entertainment: Game Development

HOWEST University of Applied Sciences

09/2024 - Present Kortrijk, Belgium

Diploma in System Programming

High School of Mathematics "Dr Petar Beron" - Varna

09/2019 - 06/2024 Varna, Bulgaria

FEATURED PROJECTS

Self-Made Game Engine

github.com/DanielAdam13/GameEngine

Built a general-purpose 2D engine from scratch in C++, along with a Bomberman clone and a Jump Point Search research created entirely on it. Used to explore standard C++ style and programming patterns.

Vulkan Renderer

github.com/DanielAdam13/VulkanRenderer

A Vulkan renderer built from scratch, focused on the lowest-level API details like synchronization and command recording.

The Last Conduit - UE5 Game

dannyboi05.itch.io/thelastconduit

Lead programmer on this Unreal Engine 5 title. Created in a group of 6 people - the strictest team-working environment I've worked in. Problem solving and communication were essential.

The Perfect Shift - Gem One for TVH (Game for Client)

dannyboi05.itch.io/the-perfect-shift

Lead programmer and GitHub coordinator on a 10-person team building a Unity product for a client, with meetings and presentations at the GemOne and TVH HQs.

SKILLS

Programming Languages

C++ C# Python Lua

Game Engines

Unreal Engine 5 Unity

Graphics Rendering

Vulkan SDL DirectX 11

Build Systems and Version Control

CMake GitHub Perforce

Art/UI Tools

Blender Photoshop WPF